



OMNIKON

NATIONAL HACKATHON 2026



01 // 06

TEAM NAME Codeandcoffee

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THEME

DISASTER MANAGEMENT / AI

PROBLEM STATEMENT

**Low-Cost Landslide Early
Warning
Using Weather & Terrain Data**

Proposed Solution

Build an affordable AI-based platform that combines rainfall, soil moisture and terrain characteristics to estimate landslide risk for a specific location.

The system converts environmental conditions into a 0–100% risk score and classifies zones as Low, Moderate, High or Critical.

An interactive risk map highlights vulnerable areas, while location-based alerts communicate elevated risk to residents and response teams.

Innovation: dynamic risk updates + explainable contributing factors + software-first architecture designed for resource-constrained regions.

WEATHER

TERRAIN

AI RISK ENGINE

RISK MAP

EARLY ALERT

DATA SOURCES

Rainfall • Soil moisture • Elevation • Slope • Temperature • Historical landslides



PROCESSING

Cleaning • Normalization • Feature engineering



ML RISK ENGINE

Random Forest / XGBoost → probability score (0–100%)



DECISION LAYER

Low • Moderate • High • Critical



OUTPUT

Interactive risk map + location-based warning + dashboard

TECH STACK

Python • Pandas • NumPy • Scikit-learn • FastAPI • React • Leaflet • OpenStreetMap •
Firebase/PostgreSQL

FEASIBILITY

Prototype can begin with public weather, terrain and historical landslide datasets.

Open-source ML, mapping and web technologies keep the software cost low.

MVP can run as a cloud/web dashboard before adding live data integrations.

CHALLENGES / RISKS

Missing or incomplete environmental data

False positives / false alarms

Limited real-time data availability

Model performance varying by region

MITIGATION

Data cleaning, imputation and validation

Combine ML output with interpretable risk thresholds

Use multiple sources where available

Validate across different regions before deployment

TARGET AUDIENCE

Local communities in landslide-prone hilly regions

Local authorities and disaster-management teams

Emergency responders and field coordinators

BENEFITS

SOCIAL — Earlier awareness, preparation and actionable warnings.

ECONOMIC — Potentially lower monitoring and disaster-response costs.

ENVIRONMENTAL — Continuous monitoring of vulnerable terrain and risk zones.

OPERATIONAL — Risk maps help responders prioritize high-risk locations.

Research areas: machine-learning-based landslide susceptibility prediction; rainfall-induced landslides; landslide early-warning systems.

Planned data sources: historical landslide records, rainfall/weather observations, digital elevation and terrain data.

Implementation references: Scikit-learn, FastAPI, React, Leaflet and OpenStreetMap documentation.

REFERENCE CHECKLIST

Machine Learning-based Landslide Susceptibility Modeling in the Dibang Valley, NE India (2026)

Rainfall Thresholds for Prediction of Landslides in Idukki, India: An Empirical Approach (2019)

Early warning system for shallow landslides using rainfall threshold and slope stability analysis (2018)

Application of machine learning algorithms in landslide susceptibility mapping, Kali Valley, Kumaun Himalaya, India (2022)

AI Methodology

AI-BASED LANDSLIDE RISK PREDICTION

INPUT FEATURES

Rainfall • Soil moisture • Slope • Elevation • Temperature • Historical occurrence

PIPELINE

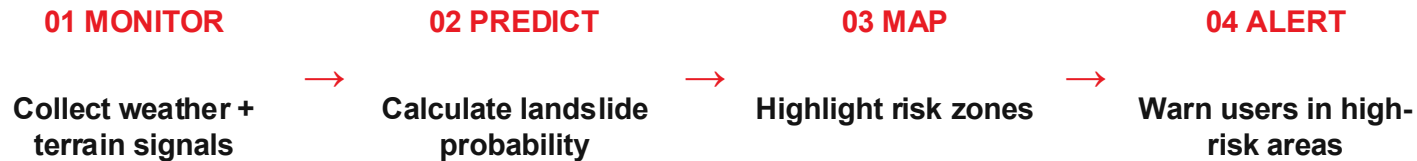
Raw data → Clean / engineer → Train & validate → Random Forest / XGBoost → Probability → Risk class

EXPLAINABLE OUTPUT

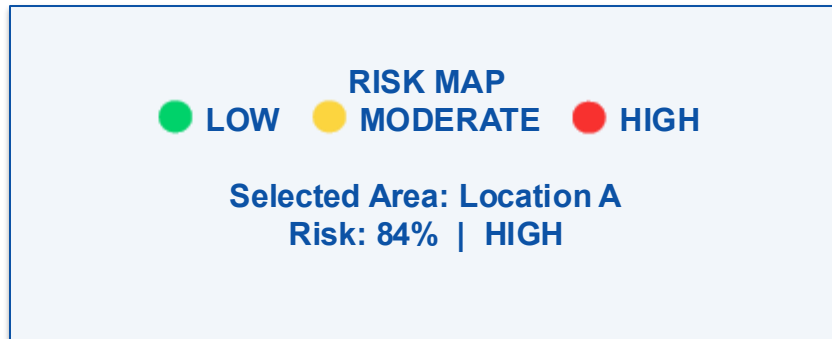
Example: 84% HIGH RISK — heavy rainfall + steep slope + high soil moisture.

PRODUCT WORKFLOW

FROM PREDICTION TO EARLY WARNING



PROTOTYPE DASHBOARD



 **WARNING**
High landslide risk detected.
Follow local authority safety
instructions.